Artificial Intelligence

ANN Properties, Advantages and Disadvantages



Lecture 04

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Underfitting and overfitting

Aspect	Underfitting	Overfitting
Training error	High	Low
Validation error	High	Higher than training
Possible response	Better features or capacity	Regularization or more data

Key ANN properties

Nonlinearity supports complex decision boundaries. Parallel distributed processing spreads computation.

Adaptivity allows weights to change from data. Generalization applies learned patterns to unseen examples.

Fault tolerance can arise from distributed representations.

Learning and memory

Knowledge is stored across weights and biases. Training changes parameters rather than program rules.

A network may memorize training data without learning a general pattern. Generalization matters more than training accuracy alone.

Advantages of ANN

Learns complex relationships from examples. Handles noisy and incomplete data. Supports classification, regression, and recognition.

Can approximate nonlinear functions. Scales to high-dimensional inputs.

Disadvantages of neural networks

Training may require substantial data, computation and experimentation.

Architecture and learning settings affect stability and performance.

Complex decisions may be difficult to explain.

Poor data can create biased or unsafe behavior.

Training accuracy alone does not establish reliable generalization.

Capacity and fitting behavior

A model with insufficient capacity may miss useful relationships.

A model with excessive capacity may fit noise in the training set.

Compare training and validation errors when selecting capacity.

Regularization and early stopping can reduce overfitting.

Model assessment checklist

Define the task and an appropriate performance metric.

Separate training, validation and test examples.

Compare performance on training and unseen data.

Inspect failure cases, noise sensitivity and relevant subgroups.

Distributed processing

Many simple units contribute to the computation.

Knowledge is distributed across weights.

Damage to one unit may not destroy the entire function.

Nonlinearity and Generalization

Nonlinear activation functions model curved relationships. Multiple layers combine simple features into complex representations.

A purely linear network cannot solve nonlinear class boundaries. A trained network should perform well on unseen samples.

Representative training data improves generalization. Validation data guides model selection.

Fault tolerance

Distributed representations can tolerate limited noise. Redundant signals reduce dependence on one feature.

Severe distribution changes can still cause failure.

Underfitting and overfitting

Underfitting: training and validation errors both remain high.

Possible responses: improve features, increase capacity or train more effectively.

Overfitting: training error is low while validation error is substantially higher.

Possible responses: more representative data, regularization or early stopping.

Application selection

Use an ANN when relationships are complex or nonlinear. Ensure enough representative examples are available.

Choose an evaluation metric that matches the real task.

Ethical and operational issues

Protect personal and sensitive data. Measure performance across relevant groups. Monitor the model after deployment.

Provide a safe fallback when confidence is low.

Worked comparison

Task A: predicting a continuous temperature requires regression. Task B: identifying a fault type requires classification.

The output layer and metric must match the task.

Benefits and operational risks

Benefits include nonlinear modeling, adaptation and distributed representations.

Tolerance of noise or missing data depends on training and architecture.

Risks include overfitting, biased outputs and changes in the input distribution.

Monitor errors after deployment and provide a fallback for unreliable predictions.

Benefits and associated risks

Property	Benefit	Limitation / risk
Adaptation	Learns from examples	Depends on data quality
Nonlinearity	Models complex relations	Can be hard to explain
High capacity	Flexible representations	Can memorize noise